PIZZA BARN

Excel Dashboard Study Handbook

Lessons 126-154 | Three Dashboard Projects

Customer Service Dashboard | Finance Dashboard | Orders Management Dashboard

A consolidated learning reference with step-by-step Excel workflows, formulas, PivotTable configurations, chart design, slicer setup, troubleshooting, and revision exercises.

Completion status: The course section discussed in our lessons is complete through Lesson 154.

Source note: This handbook consolidates the lessons developed in our conversation. Detailed procedures are drawn primarily from the completed lesson explanations. Practice numbers are illustrative unless expressly labeled as instructor observations; they should not be used as verified business results.

Contents and learning map

Module	Lessons	Main skills
1. Customer Service	126-137	PivotTables, PivotCharts, slicers, timeline, chart formatting, layout, colors
2. Finance	138-146	Date grouping, ticket-size grouping, combo charts, donut charts, theme behavior, slicer styles, insights
3. Orders Management	147-154	Calculated columns, COUNT/SUM/AVERAGE, KPI cards, chart alignment, log scale, creative shapes, multiple slicers

Standard workflow shared by all three projects

• Understand the business questions and inspect source data.

• Prepare data fields and create a dedicated Pivot worksheet.

• Build and verify PivotTables before adding charts.

• Insert PivotCharts and place them on a separate Dashboard worksheet.

• Format titles, labels, colors, axes, legends, and white space.

• Insert slicers and connect them to every compatible PivotTable.

• Test filtered totals and chart behavior; add clear business takeaways.

Common workbook structure

Worksheet	Purpose
Raw source sheet	Original records; keep clean and tabular
...-Pivot	PivotTables that drive analytical visuals
...-Dashboard	Presentation layer containing PivotCharts, slicers, KPI visuals, insights

Navigation: Each module includes a learning summary, exact Excel commands, an applied example or configuration table, and mistakes to check.

Module 1 | Customer Service Dashboard (Lessons 126-137)

The first Pizza Barn dashboard establishes the end-to-end process: translate customer-service questions into PivotTables, visualize them through PivotCharts, and combine slicers and a timeline into a usable dashboard.

Lesson progression

Lessons	Skills covered
126-129	Customer-service business questions; source inspection; PivotTable setup and basic visualization
130-133	Expand the analysis with multiple PivotCharts; adjust chart choice and readable labels
134-135	Organize dashboard charts; prepare slicer and timeline interactions
136-137	Refine background, alignment, colors, spacing, and dashboard usability

Core technique: build PivotTables from questions

• Click within the source dataset and choose Insert > PivotTable.

• Use a dedicated Customer Service Pivot worksheet.

• Drag a category such as service type, response category, or period into Rows (use actual source fields).

• Drag a relevant ID/record field into Values; choose COUNT for frequencies, SUM for additive numeric totals, or AVERAGE for mean values.

• Choose a chart that matches the question: columns/bars for comparison, lines for chronological trends, and donuts for a limited number of meaningful share categories.

Dashboard controls

• Insert Slicer: select any source-compatible PivotTable > PivotTable Analyze > Insert Slicer.

• Insert Timeline: select a compatible PivotTable > PivotTable Analyze > Insert Timeline and choose a true date field.

• Use Slicer > Report Connections to select compatible PivotTables that should respond to the slicer.

• Test the timeline at the intended granularity and verify that all charts and totals respond.

Layout and presentation checks

• Move PivotCharts from the Pivot worksheet to the dashboard (cut/paste the entire chart object).

• Hide worksheet gridlines (View > Gridlines) for a cleaner presentation.

• Remove PivotChart field buttons where possible, unnecessary chart borders, redundant legends, and excess gridlines.

• Keep chart headings readable and use consistent theme colors and spacing.

• Retain a clear dashboard title and confirm all interactions before delivery.

Practice: Explain why a slicer can change one chart but not another, and demonstrate how Report Connections resolves it.

Module 2 | Finance Dashboard (Lessons 138-146)

Lessons 138-140: PivotTable preparation and grouping

Prepare Finance source data and create Finance-Pivot and Finance-Dashboard. Build daily sales summaries and explore grouping numeric fields into ticket-size bands. Use a second Values field to compare distinct metrics from the same source.

Finance analysis	PivotTable method
Trend of Overall Sales	Sale Date in Rows; SUM of sales/revenue in Values
Trend of Average Sales	Sale Date in Rows; AVERAGE of sales/revenue in Values
Ticket-size revenue contribution	Grouped ticket-size bins in Rows; SUM of revenue in Values
Number of sales by ticket size	Grouped ticket-size bins in Rows; COUNT of sale records in Values
Product sales and average value	Product in Rows; SUM and AVERAGE added separately to Values

Lessons 141-143: Five main charts

• Line Chart: Trend of Overall Sales.

• Line Chart: Trend of Average Sales.

• Doughnut Chart: Overall Sales by Ticket Size.

• Doughnut Chart: Number of Sales by Ticket Size.

• Combo Chart: Overall Sales by Product as columns plus Average Sales as a line on a secondary axis.

For the combo chart: select chart > Chart Design > Change Chart Type > Combo, assign each series the intended type, and enable Secondary Axis for the differently scaled average if appropriate. Label both axes carefully.

Finance dashboard: Region slicer

Insert a Region slicer, arrange its four options horizontally if required, and use Report Connections to connect all five compatible PivotTables. Changing Region should update each Finance visualization.

Chart cleanup checklist

• Remove PivotChart field buttons and worksheet gridlines.

• Remove unnecessary titles, chart fills, outlines, axes, and legends only when the remaining context is clear.

• Adjust doughnut hole size (the instructor uses approximately 50%) and display category/percentage labels as useful.

• Tune column Gap Width and line width/markers where needed.

• Use dynamically linked chart titles: select the title, click the formula bar, type = and click the label cell.

Module 2 | Finance Colors, Slicer Styles and Insights

Lesson 144: Theme colors versus fixed background colors

A Home > Fill Color selection from Theme Colors can change when Page Layout > Colors changes the workbook palette. To keep the background stable, select dashboard cells > Home > Fill Color > More Colors > Standard (or Custom) and choose a fixed color. You can then change Page Layout > Colors to experiment with chart palettes without changing that fixed background.

The instructor experiments with background hues, keeps a green-themed chart appearance, and adds a centered Finance Dashboard title. Exact RGB colors are not specified.

Lesson 145: Create a custom slicer style

• Select the Region slicer > Slicer tab > Slicer Styles > More > New Slicer Style; name it Dashboard.

• Whole Slicer: remove the unwanted fill, recognizing that true transparency varies by Excel version.

• Header: apply bold type and a subtle bottom border.

• Selected Item with Data: darker fill; Unselected Item with Data: lighter green.

• Selected Item with No Data: visually distinct light/red state; Unselected Item with No Data: muted grey.

• Leave hover states unchanged in this demonstration; set the style as workbook default if wanted; save and apply it to the existing slicer.

Lesson 146: Add business insights with text boxes

Insert > Text Box, position the executive summary, choose Shape Format > Shape Fill > No Fill, and use a subtle/dashed outline if desired. Duplicate the box for chart-level observations.

Insight type	Use
Dashboard-level	Summarize key patterns across several charts; establish reporting period and major changes
Chart-level	Explain one visualization, including the metric, period, and meaningful change

The instructor describes approximately 70% of revenue from two ticket-size brackets ($500-$699 and $700-$899), a drop toward about $5,000 daily sales around late July, and possible weakness in Products 5 and 6. Treat these as transcript observations, not independently validated data.

Critical limitation: Manually typed text boxes do not automatically update when the Region slicer changes. Label their scope (for example, All Regions) or use a formula-driven summary with a linked display element.

Module 3 | Orders Dashboard (Lessons 147-154)

Lesson 147: Set up and calculate source measures

Unhide the Orders data, create Orders-Pivot and Orders-Dashboard, prepare the dashboard view, and calculate before- and after-discount revenue. Use actual field names and adjust the formulas to match the workbook.

Field / KPI	Calculation or aggregation
Revenue Before Discount	Quantity x Unit Price
Revenue After Discount	Before Discount x (1 - Discount Rate), if Discount Rate is decimal
Discount Amount (negative convention)	Revenue After Discount - Revenue Before Discount
Total Orders	COUNT of nonblank transaction records (or distinct Order ID if one order has many rows)
Total Revenue	SUM of Revenue After Discount
Average Revenue	AVERAGE of Revenue After Discount
Average Discount Amount	AVERAGE of signed Discount Amount

Formula examples (illustrative column references)

```
=ROUND(C2*D2,0)
```

```
=ROUND(F2-(F2*E2),0)
```

```
=G2-F2
```

In these examples C = Quantity, D = Unit Price, E = Discount Rate, F = Before Discount, and G = After Discount. The instructor experiments with zero-decimal ROUND; real financial reports usually need precision matching their business rules. Rounding each row can change aggregated totals.

Lesson 148: PivotTable field layout

Analysis	Rows	Columns	Values
Most Ordered Products	Product	-	COUNT and SUM of after-discount revenue
Daily Order Trend	Sale Date	-	COUNT of transactions
Daily Revenue Trend	Sale Date	-	SUM of after-discount revenue
Product-wise Revenue	Sale Date	Product	SUM of after-discount revenue

Open a missing Field List using PivotTable Analyze > Field List. If Excel groups dates into Months/Quarters automatically, remove those grouping levels when you need daily results. Refresh the PivotTable after adding new Table columns.

Module 3 | Charts, Layout and KPI Cards

Lesson 149: Create the first eight charts

Chart	Initial choice
Total Orders / Total Revenue / Average Revenue / Average Discount	Four 2-D horizontal Bar Charts
Most Ordered Products	Clustered Column (count plus revenue can need scale adjustment)
Daily Number of Sales	Line with Markers
Daily Revenue	Line with Markers (later changed to Area)
Product-wise Revenue	Column (later removed as redundant in this dashboard design)

Lesson 150: Arrange the dashboard

• Cut charts from Orders-Pivot and paste into Orders-Dashboard.

• Place four same-size KPI charts along the top. Arrange analytical charts below.

• Use Chart Format > Size for consistent height and width.

• Select objects with Ctrl > Chart/Shape Format > Align > Align Top, then Distribute Horizontally.

• Use Snap to Grid to help objects follow worksheet cell edges.

• Remove unnecessary chart and plot area fills, outlines, legends, and gridlines.

• Increase label sizes enough to be readable at dashboard zoom.

Lesson 151: Convert Bar Charts into KPI cards

• Select KPI Bar Chart > Chart Design > Add Chart Element > Data Labels > Center.

• Hide horizontal and vertical axes and unnecessary gridlines.

• Choose a readable white or high-contrast font color for the value.

• Format Data Series > Series Options > Gap Width: about 15% to make a thick horizontal bar.

• Add chart titles; optionally link titles to Orders-Pivot cells using the formula bar.

• Increase KPI label sizes (the instructor explores about 48-54 pt), avoiding clipping.

• Set count and currency formats appropriately through PivotTable Value Field Settings > Number Format.

• Experiment with a matching thick Solid Line border and Round Cap; the effect varies by Excel version.

KPI quality control: When discount amounts are negative, confirm the bar orientation and displayed sign are intentional. A very large currency number may need a smaller label or more width.

Module 3 | Advanced Charts and Interactivity

Lesson 152: Refine analytical charts

• Most Ordered Products: add Outside End labels and rename series to Number of Orders and Revenue.

• The instructor experiments with Format Axis > Logarithmic Scale to show values of different magnitudes. Log scales do not support zero/negative values and do not make currencies comparable to counts.

• Remove the redundant product revenue chart in the instructor's revised layout.

• Change the Daily Revenue Line Chart to a 2-D Area Chart; improve the outline and labels.

• Keep and resize the Daily Orders Line Chart.

• Link analytical chart titles to relevant Orders-Pivot cells.

Lesson 153: Add and connect Order Type and Agent slicers

• Select an Orders PivotTable > PivotTable Analyze > Insert Slicer. Choose Order Type and Agent fields.

• Move both slicers to Orders-Dashboard. Set Order Type to 2 columns and Agent to 3 columns where practical.

• Add Orders Management Dashboard title using top worksheet cells and Merge & Center, or a text box.

• Select each slicer > Slicer > Report Connections; check all compatible Orders PivotTables.

• Test Online/Physical and Agent selections individually and together; each linked KPI and chart should respond.

Creative chart styling

• Format Data Series > Gap Width to increase thickness of columns.

• Adjust Series Overlap for multiple series, checking that neither becomes hidden.

• Line chart: use a stronger line, built-in markers, and modest marker size (the instructor tries square markers).

• Area chart: use a contrasting boundary line for definition.

• Insert a rounded Shape, remove outline, style it, copy it, select a chart series or point, and paste to apply a picture-like shape fill in supported Excel versions.

Shape-copy techniques vary by Excel version and series type. Check whether an actual chart data series is selected rather than the floating chart object.

Module 3 | Final Slicer Styles and Dashboard Completion

Lesson 154: Reuse a custom style

• Select a slicer > Slicer > Slicer Styles > More. Right-click an existing custom style and choose Duplicate.

• Name the copy Orders Dashboard Style; modify Whole Slicer fill/border and button states.

• Use a prominent selected-item fill and subtle unselected-item fill.

• Save once and apply that style to both Order Type and Agent slicers.

• Hide an unnecessary header: right-click slicer > Slicer Settings > uncheck Display Header. Retain a separate label if needed.

Positioning versus locking

Right-click the slicer border > Size and Properties > Properties > Do not move or size with cells. This keeps the slicer anchored relative to changes in worksheet dimensions; it does **not** fully stop manual dragging. Restricting edits may require worksheet protection, which must be tested so slicer interaction still works.

Final Orders dashboard inventory

Element	What it communicates
Four KPI charts	Orders, total revenue, average revenue, average signed discount
Most Ordered Products chart	Product order count; revenue where relevant
Daily Orders Line Chart	Transaction frequency over time
Daily Revenue Area Chart	Revenue magnitude and trend over time
Order Type slicer	Online versus Physical transaction views
Agent slicer	Agent-level filtering
Titles, styling, protection settings	Readability and stable presentation

Before sharing: Refresh PivotTables, check totals under All filters, test the intersection of Order Type and Agent, review titles, verify all data labels, and save a clean default filter state.

Master Excel Command Reference

Task	Excel path / action
Create PivotTable	Insert > PivotTable
Show fields	PivotTable Analyze > Field List
Change aggregate	Right-click value > Value Field Settings
Update new records/columns	PivotTable Analyze > Refresh
Insert PivotChart	PivotTable Analyze > PivotChart; or Insert > Recommended Charts
Change chart type	Chart Design > Change Chart Type
Add title/labels	Chart Design > Add Chart Element
Format bar thickness	Right-click data series > Format Data Series > Gap Width
Format series overlap	Format Data Series > Series Overlap
Format logarithmic axis	Right-click value axis > Format Axis > Logarithmic Scale
Show / hide worksheet gridlines	View > Gridlines
Align objects	Chart/Shape Format > Align
Insert slicer	PivotTable Analyze > Insert Slicer
Connect slicer	Slicer tab > Report Connections
Create slicer style	Slicer > Slicer Styles > More > New Slicer Style
Hide slicer header	Right-click slicer > Slicer Settings > Display Header
Set slicer cell behavior	Right-click border > Size and Properties > Properties
Set fixed fill	Home > Fill Color > More Colors > Standard / Custom
Change theme palette	Page Layout > Colors
Add dashboard insight	Insert > Text Box; Shape Format > No Fill

Troubleshooting and Quality Assurance

Symptom	Likely reason / next action
Only one chart responds to slicer	Open Report Connections; select all compatible PivotTables
New calculated field missing	Check Excel Table includes the column; refresh PivotTables
Chart looks like an ordinary KPI bar	Center the label; remove axes; reduce Gap Width; enlarge label
Chart series is hard to see	Different scale or overcrowding; consider alternate chart/axis, and label units clearly
Theme changes dashboard background	Background uses a Theme Color; switch to a fixed color
Text insight contradicts filtered data	Static text box; label filter scope or use linked calculations
Slicer moves when columns resize	Set Do not move or size with cells
Slicer still draggable	Positioning setting is not protection; review worksheet protection and test
Pasted shape is floating on chart	Data series was not selected; reselect series and paste if supported
Revenue totals differ after ROUND	Rounding rows early can change sums; confirm precision policy

Course-wide publishing checklist

• Source data is complete; columns and data types are correct.

• Every PivotTable uses the intended business aggregation.

• Dashboard totals agree with PivotTable totals in the default filter state.

• Charts have clear titles, meaningful labels, and appropriate units.

• Slicer connections are complete and tested.

• Unused legend, border, axis, and field buttons are removed where appropriate.

• Text is readable at intended zoom and no chart objects overlap.

• Static narrative insights are correctly scoped to their filter context.

• Dashboard is saved in a sensible state ready for business users.

Practice and Self-Test

Hands-on tasks

• Build two PivotTables from the same source - one COUNT of orders and one SUM of after-discount revenue. Explain the difference.

• Add a numeric ticket-size grouping, then produce one revenue-share donut and one transaction-count donut.

• Create a combo chart comparing SUM of revenue and AVERAGE of transaction value by product. Label any secondary axis.

• Insert one slicer and connect it to three compatible PivotTables. Confirm each chart changes.

• Create an Orders KPI card from a single-series Bar Chart using Center labels, no axes, and a low Gap Width.

• Test fixed cell fills versus Theme Colors by changing Page Layout > Colors.

• Add an overall summary and identify why manually typed insight text can become stale when slicers change.

Knowledge-check answers

Question	Key answer
COUNT vs SUM?	COUNT measures number of nonblank records; SUM adds numeric values.
When use a line chart?	When displaying change across a meaningful ordered time sequence.
When use a secondary axis?	With care, for distinct series at materially different scales; label both clearly.
Where connect a slicer?	Slicer tab > Report Connections.
How to make KPI bar thicker?	Reduce Format Data Series > Gap Width.
Why use fixed background colors?	So workbook theme palette changes do not recolor the selected cell background.
Does Do not move or size lock a slicer?	No; it controls response to cell changes, not all user dragging.
Can a typed text box auto-update?	Not without additional linking/calculation mechanisms.

Final summary

Lessons 126-154 cover the complete three-dashboard learning section in our conversation: customer service, finance, and orders management. Together they teach Excel analysis, visualization, interactive filtering, dashboard formatting, and business communication. Use the procedures as a reference, and validate all figures against the actual source workbook before publishing.